Siddhant Tandon
Ann Arbor, MI | siddtan@umich.edu | (734) 257-0265 | siddhant-tandon.com
Australian Citizen - E3 Work Visa Eligible (No H1-B Required)

EDUCATION

University of Michigan — **Ann Arbor, MI**
Master of Science in Robotics — *May 2027*

Monash University — **Melbourne, Australia**
Master of Mechanical Engineering — *Dec 2023*
Graduated with Distinction.

Monash University — **Melbourne, Australia**
Bachelor of Mechatronic Engineering (Hons) — *Dec 2022*

WORK EXPERIENCE

NASA Jet Propulsion Laboratory (JPL) — **Pasadena, CA**
Research Intern — *May 2026 – Sep 2026*

- Led a research project assessing language models for task planning and online learning as part of the NeBula autonomy team.
- Built a high-fidelity IsaacSim benchmark and closed-loop harness in which a vision language model writes, validates and installs its own control stack at runtime, and benchmarked five models across four navigation and search tasks in 160+ simulation experiments.
- Produced a feasibility map for model authored autonomy, quantifying the best-case variance in performance that constrains the approach, helping narrow the team's future research direction. Presented findings to lab technical leaders (research products pending public release approval).

Defence Science & Technology Group (DSTG) — **Melbourne, Australia**
Robotics Research Engineer II — *Jan 2024 – Aug 2025*

- DSTG is the Australian Department of Defence's R&D agency, where I designed, built and integrated uncrewed aerial systems (UAS).
- Built custom quadcopters, and designed (CAD) and built sensor payloads for both quadcopter and fixed-wing UAS.
- Developed camera tracking and localization algorithms, AI object identification and an RF profiling toolkit for NVIDIA Jetsons aboard the quadcopters and an in-house, long-range fixed-wing UAS, delivered them to Defence clients after field testing, and earned our team the Chief Defence Scientist's Science & Engineering Excellence Award.
- Led development of a high-fidelity Unreal Engine/Python simulator for testing computer vision, control and autonomy algorithms for fixed-wing and quadcopter UAS.
- One of three evaluators on the $6M+ ASCA Sovereign UAS Program, where I designed and ran validation exercises and tests, evaluated and directed multiple contractors, and earned a Gold Commendation from the Vice Chief of the Defence Force.
- Led junior engineers on deliverables for two international research collaborations, which culminated in joint capability demonstrations in the field. Represented DSTG at overseas technical workshops and reported directly to senior embassy staff.
- Promoted early in Sept. 2024 due to outstanding performance.

Intern — *Dec 2020 – Dec 2023*

- Developed an augmented reality application (Unity/C# with ROS) enabling gaze-based control of a robotic arm, and exhibited the prototype at internal conferences to senior Defence officials and collaborators.
- Collaboratively developed a package for resource-constrained UAS navigation and coordination using bug algorithms and communications daisy-chaining.
- Full-time placement Nov. 2020 – Mar. 2022, and part-time while studying until Jan. 2024, on a STEM cadet scholarship.

Immersive Analytics Laboratory – Monash University — **Melbourne, Australia**
Research Assistant — *Nov 2019 – Nov 2020*

- One of three engineers who developed the MADE-Axis, a wireless, modular tangible controller with two actuated sliders and a rotary encoder that delivers haptic feedback for data exploration on desktop, VR and AR.
- Developed device firmware and communication protocols for wired and Bluetooth connections.
- Programmed apps in C, C# and Visual Basic so the controllers work with custom research software and tools like Adobe Photoshop.
- Designed and built 4 demo apps showcasing the device's haptic interactions, from tactile cues at individual data points to sliders that actuate to represent data ranges.
- Co-authored the resulting paper (DOI: 10.1145/3488546), published at ACM ISS 2021 and awarded Best Paper Runner-up.

PROJECT EXPERIENCE

Multi-Agent Search and Exploration – SSI & CURLY Labs, University of Michigan **Ann Arbor, MI**
Graduate Researcher *Jan 2026 – Present*

- Developed a Python-based multi-agent autonomy stack integrated with Isaac Sim, enabling simulation of heterogeneous robot teams.
- Leading a project investigating novel algorithms for multi-robot search and exploration tasks. Workshop paper (1st author) accepted to RSS 2026 OWN workshop. Pre-print link here: https://tinyurl.com/yckxbruy.

DARPA APSU Challenge – CURLY Lab, University of Michigan **Ann Arbor, MI**
Graduate Researcher *Aug 2025 – Feb 2026*

- Developed multi-camera mapping and task decomposition modules in Python for competition submission.
- Designed and implemented a ROS interface integrating agents with IsaacSim and Habitat environments; led Docker containerization and submission of competition milestones.

Explainable Shared Autonomy in Drone Landings Using Augmented Reality **Melbourne, Australia**
Honors Student *Jan 2022 – Dec 2022*

- Developed an AR application to improve explainability of an AI assistant for landing drones.
- Built the drone and conducted a user study, showing that AR significantly enhanced transparency and accuracy while reducing cognitive load, though with a trade-off in efficiency. Thesis is available here: https://tinyurl.com/4yr5pk2v. The thesis was awarded a High Distinction (highest Australian grade).

OTHER EXPERIENCES

University of Michigan **Ann Arbor, MI**
Graduate Student Instructor – ROB311 (How to build robots and make them move) *Jan 2026 – Present*

Monash University **Melbourne, Australia**
Teaching Associate (Instructor) – First Year Engineering Design *Jan 2023 – Jul 2023*

Monash University **Melbourne, Australia**
Mentor & Assessor – Second Year Engineering Design *Jan 2021 – Jul 2021*

Monash High Powered Rocketry **Melbourne, Australia**
Avionics & Control Team Member *Jan 2018 – Jan 2022*

SKILLS

- *Programming Languages:* Python, C, C#, MATLAB & JavaScript/Web Development.
- *Frameworks/Software Development Competencies:* ROS, Unity, Unreal Engine, IsaacSim, Docker, Linux CLI, ArduPilot, AR/VR dev., Git, LLM interfaces (Ollama, vLLM etc.) & computer vision using OpenCV & YOLO.
- *Hardware:* Haptic & actuated interface design, sensor payload design & integration, Uncrewed Aerial System design, CFD, SolidWorks/Onshape CAD & electrical prototyping.

SCHOLARSHIPS & AWARDS

Australian Defence Force – Gold Commendation for Achievement *2025*
Vice Chief of the Defence Force (Australia)

Chief of Division Award (one team per year in a ~1,000-person division) *2025*
Chief of Platforms Division – Defence Science & Technology Group (Australia)

Science & Engineering Excellence Award (our team of 6, from a ~3,500-person organization) *2024*
Office of the Chief Defence Scientist (Australia)

STEM Cadetship Scholarship (~US$85,000) *Jan 2022 – Dec 2024*
Department of Defence (Australia)

Master's Pathway Scholarship (~US$21,000) *Jan 2018 – Dec 2023*
Monash University

Summer Research Scholarship (~US$2000) *Nov 2019 – Feb 2020*
Monash University

Dean's Honor Roll *Jan 2019, Jan 2022*
Monash University – Faculty of Engineering